Composable Exact Bounds: A Self-Configuring Hybrid of PCA and Triangle-Inequality Pruning

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ABSTRACT

Exact nearest-neighbor search can be accelerated by cheap lower bounds that prune candidates without computing full distances. Our prior work used bounds from truncated PCA projections, which are tight precisely when variance concentrates in few dimensions. We observe that such bounds are one member of a *family*: any region that provably contains a set of points yields an exact lower bound via the query-to-region distance, and different region shapes are tight for different data structures. We compose two complementary families—PCA truncation (tight for linearly correlated data) and triangle-inequality ball bounds around k-means centroids (tight for clustered data, and exact for *any* centers, so approximation quality affects only speed, never correctness)—into a single exact cascade. Because neither family helps on all data, the index *configures itself*: a three-tier diagnostic measures a sample's eigenvalue decay, detects structure beyond the covariance with two cheap statistics, and settles the question with a measured trial of the real cascade. We report results in hardware-independent work units—full-vector fetches, total bytes touched (charging the pruning machinery itself), and resident RAM per vector—because wall-clock proved regime-dependent: the same algorithm measured 4.7×, 3.6×, 2.7×, and parity under changes of numeric type, machine load, and implementation tuning, while its work counts never moved. On SIFT1M with official queries, the cascade computes full distances for 0.06% of the dataset and keeps 8–40 bytes per vector resident in RAM (versus 512 for a flat scan); on clustered data with a flat eigenvalue spectrum, the ball level prunes to 3% before a single projection is computed (13.9× wall-clock versus 2.1× for PCA alone); on SIFT the ball level's bound work cancels its savings—a verdict the diagnostic correctly predicts from a sample, protecting the practitioner from building a useless level. Exactness is verified against ground truth on every query. All code, diagnostics, and experiments are public.

1 INTRODUCTION

Finding the exact nearest neighbor in a large, high-dimensional dataset is bottlenecked by the cost of computing full distances. Approximate methods (LSH, HNSW [10], IVF [11]) trade recall for speed; for applications where a missed true neighbor is unacceptable, the alternative is exact pruning: compute a cheap *lower bound* on each candidate's distance, and discard the candidate if the bound already exceeds the best distance found so far. Our prior work [1] built such an index from truncated PCA projections: the distance between k-dimensional PCA prefixes lower-bounds the true distance, so a cascade of progressively finer prefixes prunes over 99.9% of the SIFT1M dataset while returning provably exact results.

That bound has a blind spot, and it is structural. PCA truncation is tight exactly when variance concentrates in few principal directions—linearly correlated data. Datasets whose structure is *clustered* rather than correlated can present a flat eigenvalue spectrum, collapsing the PCA cascade toward brute force, while remaining highly compressible in a different sense: points huddle near a modest number of centroids. No single bound family serves all data, and, worse, a practitioner has no way to know before building an index which family—if any—their data supports.

This paper makes two moves. First, **exact bounds compose**: the maximum of valid lower bounds is a valid lower bound, so bound families with complementary blind spots can share one cascade. We add a triangle-inequality *ball bound* around k-means centroids—the clustered-data complement of PCA truncation—and show that it is exact for *any* choice of centers: clustering quality affects only pruning power, never correctness. Second, **the composition is chosen by measurement**: a three-tier diagnostic, run on a sample in seconds, rates the PCA levels from the eigenvalue decay, detects structure beyond the covariance with two cheap statistics, and—because we show statistics alone mispredict in both directions—settles the question with a measured trial of the real cascade, emitting an ADD/SKIP verdict per level.

There is also an economic motivation. An exhaustive flat scan requires the entire dataset resident in RAM—the most expensive byte in the machine, running roughly an order of magnitude above NVMe per byte, and more under supply pressure. The pruning cascade needs only its metadata resident: 8–40 bytes per vector versus 512 bytes per vector for `fp32` data. At one billion vectors this is 8–40 GB of RAM versus 512 GB—a commodity box versus a specialty one—with exactness preserved.

Our contributions:

1. A **containing-region principle** unifying exact pruning bounds, with the ball bound as the clustered-data complement to PCA truncation (§3).
2. A **composed exact cascade**—ball level plus PCA levels—with a radius re-probe that keeps the cascade as tight as PCA-only. Storage overhead is ~8 bytes per point (§5).
3. A **three-tier self-configuration funnel**: the statistics nominate, the measurement decides; negative verdicts included (§4).
4. **Reproducible evidence in hardware-independent work units**, with wall-clock demoted to validating a cost model, plus negative controls and a sample-to-full-scale extrapolation check (§6).

Table 1: Shape families and their exact bounds. Each is tight for a different data structure; each mirrors a quantizer family.

Shape	Bound cost	Tight for	Quantizer twin
ball	$\mid \|q{-}c\| - d_p \mid$	clusters	VQ
ellipsoid	per-axis scaled	stretched clusters	OPQ
slab	projection residual	local low rank	local PCA
cone×annulus	closed form	radial/gain data	gain–shape

2 BACKGROUND AND RELATED WORK

Paper I in brief. For data rotated onto its principal axes, the squared distance restricted to the first k coordinates is a lower bound on the full squared distance (the discarded terms are non-negative). The algorithm of [1] exploits this with a cascade of prefix widths $(8, 16, 32, 64)$ and a dynamic radius: a Top-50 probe at the coarsest level computes exact distances for the 50 most promising candidates, seeding a mathematically safe pruning radius without prior knowledge. On SIFT1M the cascade discards all but an average of 564 of 1,000,000 vectors before any full-dimensional evaluation.

Lineage. Neither bound family is new, and we do not claim otherwise. Lower-bounding by dimensionality reduction descends from the GEMINI framework [2] and the VA-file [3]. Pruning by the triangle inequality around reference points is the basis of ball trees [7], vantage-point trees [8], GNAT [9], and Elkan's accelerated k-means [6]. The approximate cousins are FAISS IVF [11] (the same k-means geometry without exactness) and OPQ [5] (a per-cell rotation, the ellipsoid rung of the shape ladder in §3). Disk-resident ANN systems such as DiskANN [12] and SPANN [13] target the same memory-economics regime we do, but are approximate. Our contribution is the *exact composition* of the two bound families in one cascade, the *measured self-configuration*, and a diagnostic that also tells the practitioner when to walk away.

3 COMPOSABLE EXACT BOUNDS

3.1 The Containing-Region Principle

Let $R \subseteq \mathbb{R}^d$ be any region that provably contains a set of points C. Then for every $p \in C$ and any query q,

$$\|q - p\| \;\geq\; \mathrm{dist}(q, R), \tag{1}$$

so $\mathrm{dist}(q, R)$ is an exact pruning bound for every member of C at once. A bound family is therefore a *shape* family, and a shape is admissible when (a) containment is guaranteed and (b) $\mathrm{dist}(q, R)$ is cheap. Table 1 lists the natural ladder. Each geometric family is the twin of a quantizer family: a quantizer's codeword-plus-distortion model *is* a containing region. We use the first two rungs; the others are available under the same principle.

3.2 The Ball Bound

Cluster the dataset with k-means. For each point p assigned to centroid c, store the scalar $d_p = \|p - c\|$ once, offline. At query time, compute the k centroid distances $\|q - c_j\|$ (with $k \ll N$), and the triangle inequality gives, for every point,

$$\|q - p\| \;\geq\; \big|\, \|q - c_{a(p)}\| - d_p \,\big|, \tag{2}$$

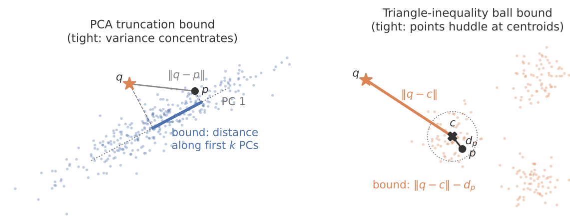

Figure 1: The two exact bound geometries. Left: the PCA truncation bound is the distance between the projections onto the first k principal components—tight when variance concentrates along few directions. Right: the ball bound $\|q - c\| - d_p$ from the triangle inequality—tight when points huddle near their centroids. Each is loose exactly where the other is tight.

where $a(p)$ is p's cluster. Evaluating (2) for all N points requires one gather and one subtraction per point—*no distance computations*. A coarser, per-cluster form uses the cluster radius $R_j = \max_{p \in j} d_p$: if $\|q - c_j\| - R_j$ exceeds the current radius, the entire cluster is eliminated in one comparison, which in an out-of-core layout skips a whole contiguous block (§5.3).

3.3 Exactness Is Unconditional; Quality Is Not

Inequality (2) is the triangle inequality: it holds for *any* reference points whatsoever. k-means enters only because it minimizes the mean d_p^2, which is what makes the bound *tight*. Three corollaries follow. Stale centers remain correct as the dataset drifts (pruning degrades gracefully; re-cluster when a diagnostic says so). Insertion is $O(1)$: assign the new point to its nearest existing centroid and store d_p; no re-clustering is required for correctness. And the composed index inherits Paper I's property in mirror image: there, any orthogonal rotation is exact and PCA makes it tight; here, any centers are exact and k-means makes them tight. Since the maximum of lower bounds is a lower bound, the two families compose freely in one cascade.

3.4 Complementarity

PCA truncation is tight iff the eigenvalue spectrum decays steeply—a *global, linear* property (Figure 1, left). The ball bound is tight iff points sit close to centroids relative to the spread of $\|q - c\|$—a *local* property that holds for multimodal data and, more generally, for data of low intrinsic dimension, where d_p and the centroid distances spread widely instead of concentrating. The blind spots are disjoint by construction. Our clearest demonstration: a synthetic dataset of 512 tight clusters whose centers span all dimensions has a nearly flat spectrum (95% of variance needs 111 of 128 dimensions; the PCA cascade manages 2.1×), yet the ball bound prunes to 3% before a single projection is computed.

4 THE SELF-CONFIGURATION FUNNEL

The funnel answers, from a sample and in seconds, "which levels should this index have for *my* data?"

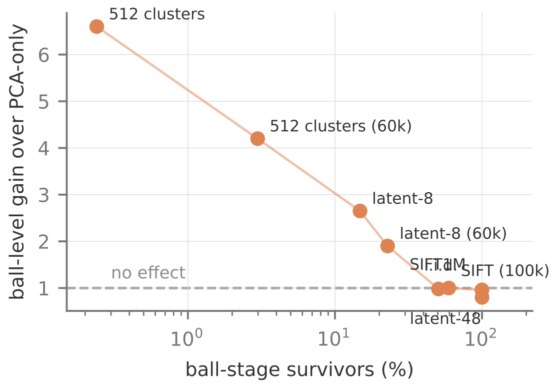

Figure 2: The measured ball-value law across all eight datasets and trials (parity code): the ball level's wall-clock gain over the PCA-only cascade is a monotone function of its own pruning rate. The funnel's tier-3 trial measures precisely the x-axis quantity, which is why it predicts full-scale outcomes from a sample—including the negative verdicts below the dashed line.

Tier 1 — eigenvalue decay. Fit PCA on a sample; the cumulative explained variance at each cascade width is the expected tightness of that PCA level. This tier rates the PCA cascade (and reproduces Paper I's SIFT1M behavior from a 100k sample).

Tier 2 — beyond-covariance statistics. Two cheap statistics on the PCA-rotated sample: the mean excess kurtosis of components (marginal non-Gaussianity), and sqdep, the mean $|\mathrm{corr}(z_i^2, z_j^2)|$ over component pairs (joint dependence that no rotation can remove; finite-sample noise floor $\approx \sqrt{2/n}$). These detect that structure beyond the covariance *exists*—not whether it is *harvestable*.

Tier 3 — the measured trial. Fit the real hybrid index on the sample, run the real cascade with the ball level on and off, and report the measured ratio and the ball stage's own pruning rate. The verdict is ADD or SKIP.

Tier 3 must be the decider, and the evidence is that the statistics mispredict in *both* directions. On SIFT, tier 2 fires (sqdep = 0.032, 7× the floor: real beyond-covariance structure) yet the trial says SKIP (1.0×): ball bounds prune only 50–59%, and the bound work cancels the savings. On a latent-dimension-8 Gaussian, tier 2 is clean (kurtosis ≈ 0, sqdep at floor) yet the trial says ADD (1.9×): low intrinsic dimension spreads d_p widely, so the bounds bite (pruning to 23%) with zero multimodality. Across all datasets the measured value of the ball level is a monotone function of its own pruning rate—survivors of 3% ⟶ 4–14×; 15–23% ⟶ 2–3×; 50–59% ⟶ parity; 100% ⟶ 0.8× (a measured loss)—and the trial measures exactly this quantity (Figure 2). No statistic substitutes for it.

5 THE HYBRID ALGORITHM

5.1 Offline

Fit PCA (as in Paper I) and k-means with $k = \mathrm{clip}(\sqrt{N}, 64, 4096)$ (MiniBatch; 24 s on SIFT1M for k=1024). Store per point a cluster id (`int32`) and d_p (`float32`)—8 bytes—plus the $k \times d$ centroids and per-cluster radii.

5.2 Query Pipeline

(1) k centroid distances ($k \cdot d$ operations).
(2) Ball bounds (2) for all N points: one gather and one absolute difference each.
(3) **Probe**: exact distances to the PROBE= 50 points with the smallest ball bounds seed a safe initial radius r (Paper I's Top-50 probe, seeded by ball bounds instead of coarse projections).
(4) Ball pruning: retain points with bound$^2 < r^2$.
(5) PCA cascade (8, 16, 32, 64) on the survivors, with a **re-probe** at the first PCA stage: before pruning, exact distances to the 50 best 8-dimensional candidates tighten r. Without this step the ball-seeded radius is loose and the cascade degrades (8-D survivors 27.2% versus 14.6% with it); with it, the cascade matches PCA-only tightness stage-for-stage.
(6) Full-distance check on the final survivors. The result is provably identical to brute force, and is additionally verified against the official ground truth in every experiment.

5.3 Out-of-Core Layout

Pruning fragments I/O only if the layout ignores it—the classic index-versus-scan crossover: 600 scattered 512-byte reads lose to a full sequential scan on seek-bound media. Measured on SIFT1M (k=1024, ~488 KB blocks): the cluster-level bound alone is a weak block-skipper (40.4% of clusters survive with a 95th-percentile radius and a 5% escape set; 52.3% with the max radius), *but* the final ~600 survivors concentrate in a median of 44 of 1024 clusters. A cluster-major layout therefore converts the fetch into ~44 contiguous reads (~21 MB). The resulting two-tier design: (1) scan the small, sequential metadata—`assign`+d_p is 8 MB, and optionally the 8-dimensional PCA sketch, 32 MB, i.e. 64×/16× smaller than the data; pruning decisions never touch full vectors—then (2) fetch survivors as batched random 512-byte reads on NVMe (~0.3 MB) or as the ~44 cluster blocks on seek-bound media. Back-of-envelope: NVMe ~15 ms versus 146 ms for a full scan (~10×); HDD ~0.55 s versus 3.4 s (~6×), where the naive scattered layout would *lose* (~4.8 s). This is the wisdom of the VA-file [3], SPANN [13], and DiskANN [12]: sequential sketches, layout-aware fetches. The same k-means that supplies the bound supplies the layout.

6 EVALUATION

Metrics policy. Primary claims are stated in hardware-independent work units, because wall-clock proved regime-dependent: the same algorithm measured 4.7×, 3.6×, 2.7×, and parity under changes of numeric type, machine load, and implementation tuning, while its work counts never moved (details below). The primary metrics are (1) full-vector fetches per query (the out-of-core I/O currency), (2) total bytes touched per query, *charging the pruning machinery itself*, (3) resident RAM bytes per vector, and (4) the survivor cascade, which is the algorithmic fingerprint. Wall-clock appears as validation of a first-order cost model ($t \approx$ bytes/bandwidth +

Table 2: SIFT1M, 100 official queries, medians of 5 repetitions, implementation parity. All methods return the ground-truth neighbor on every query.

Method	ms/query	Speedup	Survivors per stage (%)
brute (NumPy)	167.3	1.0×	–
PCA-only [1]	61.1	2.7×	14.7 / 5.2 / 0.8 / 0.06
hybrid	62.2	2.7×	ball 50.6 → 14.6 / 5.1 / 0.7 / 0.05
FAISS-flat, 1 thread	29.3	5.7×	(scans 100%)
FAISS-flat, batch-100	5.9	28×	(scans 100%)

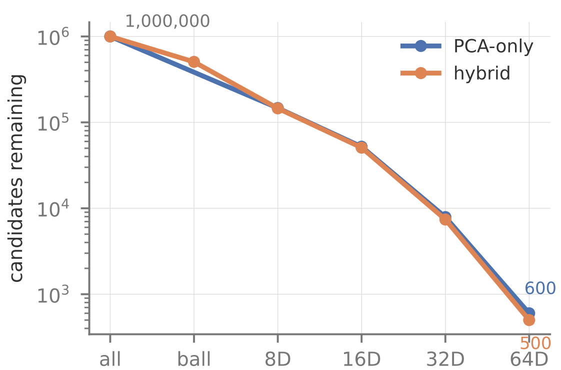

Figure 3: Candidates remaining after each stage on SIFT1M (log scale, average over 100 official queries). The cascade discards four orders of magnitude before any full-distance evaluation; the hybrid enters the PCA stages with half the population and finishes equally tight. This profile is the algorithmic fingerprint: it was invariant under every change of numeric type, machine load, and implementation tuning in this paper.

ops/throughput), with FAISS supplying the tuned-constants reference—so a reader can plug their machine's constants into our work numbers and predict their own outcome.

Setup. Intel i9-9900K (8C/16T, AVX2+FMA, no AVX-512), dual-channel DDR4; `faiss-cpu` 1.15 dispatching its AVX2 build; NumPy and OpenBLAS at the AVX2 tier—baseline and FAISS use the same instruction set. Timings are medians of 5 repetitions on a quiet machine with warm cache, exactness verified against the official ground truth on every repetition. Results are CPU-governor-insensitive (balanced versus performance profiles: all medians within 2%). Spreads are ≤2% except where noted.

6.1 SIFT1M

Table 2 gives the in-RAM picture, and Figure 3 the cascade behind it. The PCA cascade alone reaches 2.7×; **the ball level is a wall-clock tie on SIFT**—its 50% pruning saves about what its bound computation costs—exactly as the sample trial predicts (SKIP, §4). The hybrid's SIFT-side value is therefore out-of-core and residency only (§5.3); the wall-clock case for the ball level rests on clustered and low-intrinsic-dimension data (§6.2). The survivor cascade reproduces Paper I exactly.

Table 3: Hardware-independent work per query on SIFT1M. Terms are per-dimension $(x-y)^2$ evaluations; bound ops are per-point ball-bound evaluations, charged separately. Counts derive analytically from the survivor statistics (`count_ops` in the repository)—an exhaustive scan is $N \cdot d$ terms by definition, so no instrumentation of compiled libraries is needed. These columns are identical on every machine and every run.

Method	Terms (M)	Bound ops (M)	Terms vs brute
brute / FAISS-flat	128.0	–	1.0×
PCA-only [1]	12.60	0	**10.2×**
hybrid	8.67	1.00	14.8×

Table 3 states the same experiment in the primary currency. Three things the wall-clock hid become explicit. First, the cascade's *algorithmic* reduction is 10.2×—machine-free, and invariant across every run in this paper, where the millisecond columns wobbled with load. Second, FAISS-flat performs 10–15× more work than the cascade and still wins in-RAM wall-clock: the engineering-versus-algorithm separation is now quantitative rather than rhetorical. Third, the hybrid cuts terms further (12.6 → 8.7 M) but adds 1.0 M bound operations of a different mix (gathers rather than streaming reads)—which is why its wall-clock ties PCA-only despite the smaller term count, and why work accounting that omits the pruning machinery would mislead.

FAISS `IndexFlatL2` wins the in-RAM race outright, scanning all 10^6 vectors in 29.3 ms single-threaded. The gap is fusion, not SIMD: both sides run AVX2, but FAISS's kernel makes one pass (~512 MB of traffic per query) where NumPy's expression materializes temporaries (~3 passes, 1.5–2 GB), a 5.6× advantage at identical work. The comparison scopes the claim rather than defeating it: the cascade touches 1,700× fewer full vectors, requires 8–40 resident bytes per vector against 512 for any flat scan, and its bounds compose with FAISS-grade kernels rather than competing with them (the cascade could drive such a kernel over its survivor sets). Disk-resident ANN systems occupy the same economics but surrender exactness.

Why wall-clock is the wrong headline (measured). Two confounds were caught only by interrogating ratio changes. *Numeric type*: sklearn's PCA silently outputs `float64`, whose bandwidth-bound brute baseline is 1.86× costlier; a 2 × 2 decomposition on identical code measures the same cascade at 3.6× (f64—Paper I's regime, matching its reported 4.3×) versus 2.7× (f32). *Implementation parity*: an unnecessary full-array gather in our own PCA-only path inflated the ball level's apparent contribution to 2.0×; fixing it revealed the tie in Table 2. Machine load adds a further multiplier: background bandwidth contention slows the scan-bound baseline most, flattering all pruners. Survivor counts were invariant through every one of these changes. Hence the metrics policy.

Table 4: Structure regimes (synthetic, n = 200k, d = 64 for the first two; n = 60k, d = 128 for the funnel rows), implementation parity. "Ball survivors" is the ball stage's own pruning rate—the quantity the funnel's trial measures.

Dataset	PCA-only	Hybrid	Ball surv.	Verdict
512 tight clusters	2.1×	**13.9×**	3%	ADD
latent-8 Gaussian	2.1×	5.6×	15–23%	ADD
SIFT1M	2.7×	2.7×	50–59%	SKIP
latent-48 Gaussian	1.9×	0.8× of PCA	100%	SKIP
i.i.d. Gaussian	0.3×	—	100%	SKIP (all)

6.2 Structure Regimes

Table 4 spans the design space, negative results included deliberately. On the flat-spectrum clustered dataset—PCA's blind spot—the ball level prunes to 3% before any projection is computed and the hybrid reaches 13.9×; its bound requires no multiplications, so it undercuts even the 8-dimensional scan. On the latent-48 blob the ball level is measured to *hurt* (0.8×: 100% survivors, pure overhead), and on i.i.d. noise the diagnostic correctly reports that nothing works and brute force or ANN should be used. The i.i.d. row also shows the PCA cascade itself losing to brute force (0.3×)—the honest boundary of the entire approach.

6.3 Extrapolation

The funnel's trial on a 100k SIFT sample returns 1.0× (SKIP); the full-1M parity benchmark measures exactly that (61.1 versus 62.2 ms). The diagnostic predicts full-scale outcomes from a sample—including the negative verdict, which is its most valuable behavior: it prevents building a useless level.

7 LIMITATIONS

High *intrinsic* dimensionality defeats every cheap bound: distance concentration flattens the eigenvalue spectrum and the spread of d_p simultaneously, and the funnel exists precisely to detect this and say so (Table 4, last row). In-RAM wall-clock at native-code constants belongs to fused exhaustive scans (FAISS) on this dataset scale; our wall-clock case is confined to data the ball level can prune hard, and to out-of-core and RAM-constrained regimes, which we argue from measured work counts and a cost model rather than end-to-end I/O runs (a cold-cache wall-clock run is future work). The current implementation is 1-NN; top-k generalizes the radius (r = current k-th best) but is unimplemented. Ball-bound quality depends on the sample used for k-means in the same way any learned structure does, though correctness never does.

8 CONCLUSION

Exact lower bounds are shapes; shapes compose; and the right composition is a property of the data, measurable on a sample before any index is built. The resulting artifact—a hybrid of PCA truncation and triangle-inequality ball pruning that configures itself, states its results in hardware-independent work units, and honestly reports where it does not help—extends our prior exact-search work to clustered data, cuts resident RAM by 12–64×, and is available, with every experiment in this paper, at the repository below. Measure the data to choose the index; count the work to predict the time.

Artifact. Code, diagnostics, and all experiment scripts:
`github.com/taiwuchiang-gmail/hierarchical-pca-search`

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